SOLUTIONS

EE3-03

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2015

EEE/ISE PART III/IV: MEng, BEng and ACGI

COMMUNICATION SYSTEMS

Tuesday 9 December 2014, 9:00 am

Time allowed: 3:00 hours

There are 23 questions on this paper.

Answer ALL questions.

The multiple choice questions together account for 40% of the marks.

Answers to multiple choice questions 1-10 should be given on the paper itself.

Students are not permitted to use more than one answer book.

Students are not permitted to take the question paper away.

The following are provided:
A table of Fourier transforms
A Gaussian Tail Function graph

Examiners responsible:

First Marker(s): A. Manikas
Second Marker(s): B. Clerckx

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PART-I

1. Consider two analogue signals x(t) and y(t). The signal x(t) has a uniform amplitude probability density function, i.e.

$$\text{pdf}_\text{x}(x) = \frac{1}{3}\text{rect}\left\{\frac{x}{3}\right\}$$

while the signal y(t) has a Gaussian amplitude probability density function $\text{pdf}_\text{y}(y)$ of mean μ_y and standard deviation equal to $\sqrt{P_x}$where P_x denotes the average power of x(t).

Which of the following is correct? [4 marks]

(a) the average power P_x of the signal x(t) is $\frac{1}{9}$,

(b) the differential entropy H(x) of x(t) is 1.8396,

(c) the differential entropy H(y) of y(t) is 1.8396,

(d) the entropy power of x(t) is 0.52698, [x]

(e) the entropy power of y(t) is 0.52698.

2. For a speech signal transmitted using a uniform quantiser of 256 levels the bit rate at the output of the source encoder is [4 marks]

(a) 8 kbits/s,

(b) 16 kbits/s,

(c) 32 kbits/s,

(d) 64 kbits/s, [x]

(e) 128 kbits/s.

3. A digital communication system having an energy utilisation efficiency (EUE) equal to 30, operates in the presence of additive white Gaussian noise of double-sided power spectral density $\text{PSD}_n(f)$=0.5×10^{-6} W/Hz. If the channel capacity C is 16 kbits/s and the data bit rate r_b is 2 kbits/s, then the channel bandwidth B is [4 marks]

(a) 4 kHz, [x]

(b) 8 kHz,

(c) 16 kHz,

(d) 32 kHz,

(e) 64 kHz.

4. An analogue message signal $g(t)$ of maximum frequency F_g is sampled at the Nyquist rate and then quantized to $Q = 16$ levels. A source encoder encodes the output with $\log_2 Q$ bits per level. Then a digital modulator transmits each binary digit through a bandlimited channel, as one of two energy signals in the presence of white Gaussian noise of double-sided power spectral density 10^{-5} Watts/Hz. The energy E_b necessary to satisfy the condition $\mathrm{SNR}_{in-mb} = 30\mathrm{dB}$ is [4 marks]

   (a) 3×10^{-3},
   (b) 2.5×10^{-3}, [x]
   (c) 2×10^{-3},
   (d) 1.5×10^{-3},
   (e) none of the above.

5. Consider the bipolar line code waveform $\mathrm{b(t)} = \sum_{n=-\infty}^{\infty} \mathrm{a}[n].c(t - nT_{cs})$ where the autocorrelation function of the code sequence $\{\mathrm{a}(n)\}$ is as follows:

$$R_{\mathrm{aa}}(k) = \begin{cases} 1/2 & \text{if } k = 0 \\ -1/4 & \text{if } k = 1 \\ 0 & \text{if } k \geq 2 \end{cases}$$

   If $\Pr(0) = \Pr(\pm 1) = 0.5$ and $c(t) = \mathrm{rect}\left\{\frac{t}{T_{cs}}\right\}$ then the power spectral density of b(t) is [4 marks]

   (a) $T_{cs}.\mathrm{sinc}^2\{fT_{cs}\}$,
   (b) $T_{cs}^2.\mathrm{sinc}^2\{fT_{cs}\}$,
   (c) $T_{cs}.\mathrm{sinc}^2\{fT_{cs}\}.\sin^2(\pi T_{cs})$, [x]
   (d) $T_{cs}^2.\mathrm{sinc}^2\{fT_{cs}\}.\sin^2(\pi T_{cs})$,
   (e) none of the above.

6. A FH/SSS has a data rate of 4800 bits/second and uses 4 hops per data bit. What is the bandwidth if there are 1024 hopping frequencies? [4 marks]

   (a) 19.2 kHz.
   (b) 4.096 MHz.
   (c) 4.915 MHz.
   (d) 19.661 MHz. [x]
   (e) None of the above.

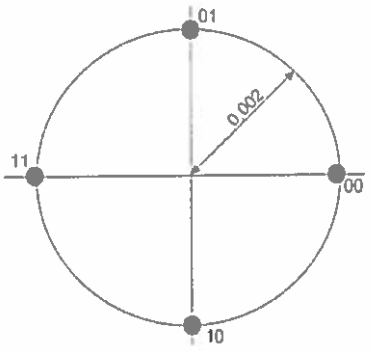
7. The minimum chip rate needed by a DS-BPSK CDMA system to resolve a multipath, with an additional path length of 15m compared to the direct path, is [4 marks]

   (a) 10 Mchips/sec, [x]
   (b) 20 Mchips/sec,
   (c) 40 Mchips/sec,
   (d) 60 Mchips/sec,
   (e) none of the above.

8. Consider a CDMA system of 256 users where each user has a protection probability equal to 4×10^{-2} and an Anti-jam Margin of 30 dB. Each user employs a feedback shift register of 21 stages, whose feedback connections are described by a primitive polynomial. The system is perfectly power controlled and the received power from each user is equal to $P = 0.1915$W operating in the presence of additive white Gaussian noise of double sided power spectral density 0.5×10^{-6} Watts/Hz. The average energy per bit E_b is [4 marks]

   (a) 0.94×10^{-3},
   (b) 1.881×10^{-3}, [x]
   (c) 2.33×10^{-3},
   (d) 4.46×10^{-3},
   (e) none of the above.

9. The constellation diagram of a QPSK Communication system is shown below



   If the bit rate is 40kbits/sec what is the amplitude of this M-ary set of signals? [4 marks]

   (a) $\sqrt{8}$mV.
   (b) 0.16V.
   (c) 0.4 V. [x]
   (d) 0.8 V.
   (e) None of the above.

10. A speech signal having a maximum frequency of 4kHz is sampled at twice the Nyquist rate and then fed through an 8-bit uniform quantizer. The generated binary sequence is then fed through a binary PSK direct sequence spread spectrum system which operates in the presence of a jammer of power 1.6 Watts and in the presence of additive white Gaussian noise with double-sided power spectral density 0.5×10^{-12} Watts/Hz. The amplitude of the BPSK signal is 0.5V.

    For this system, the spread spectrum bandwidth B_{ss} is 32 MHz and the system is not synchronised.

    The power of the jammer at the output of the despreader is [4 marks]

    (a) 1.6 mW,

    (b) 3.2 mW,

    (c) 6.4 mW, [x]

    (d) 12.8 mW,

    (e) none of the above.

PART-I SOLUTIONS

Question	Answer
1	D
2	D
3	A
4	B
5	C
6	D
7	A
8	B
9	C
10	C

PART-II

11. A binary digital communication system, having an energy utilisation efficiency (EUE) equal to 40, operates in the presence of additive white Gaussian noise of double-sided power spectral density PSD(f)=0.5106 W/Hz. If the channel capacity C is 16 kbits/s and the channel bandwidth B is 4 kHz, estimate

    (a) the noise power at the channel output; [1 marks]

    (b) the bit rate r_b. [3 marks]

Answer

$EUE = 40$

$\frac{N_0}{2} = 0.5 \times 10^{-6} \Rightarrow N_0 = 10^{-6}$

$C = 16 \frac{\text{kbits}}{\text{sec}}$

$B = 4\ \text{kHz}$

$P_n = N_0 \cdot B = 10^{-6} \times 4\text{K} = 4\text{mW}$

$\Rightarrow C = B \log_2(1+SNR_{in})$

$\Rightarrow SNR_{in} = 2^{C/B} - 1$

$\Rightarrow \frac{EUE}{BUE} = 2^{C/B} - 1$

$\Rightarrow BUE = \frac{EUE}{2^{C/B}-1}$

$\Rightarrow \frac{B}{r_b} = \frac{EUE}{2^{C/B}-1}$

$\Rightarrow r_b = \frac{(2^{C/B}-1) \cdot B}{EUE} = 1.5 \frac{\text{kbits}}{\text{sec}}$

12. Consider a random binary sequence of 0's and 1's, which are statistically independent with $\Pr(1) = \Pr(0) = 0.5$. This binary sequence is transmitted as a random signal with 1's and 0's being sent using the pulses $s_1(t)$ and $s_0(t)$ described below:

$$0 \longmapsto s_0(t) = -3 \times 10^{-3} \times \Lambda\left\{\frac{t}{1ms}\right\} \text{ Volts}$$

and

$$1 \longmapsto s_1(t) = +3 \times 10^{-3} \times \Lambda\left\{\frac{t}{1ms}\right\} \text{ Volts}$$

(a) Find the Power Spectral Density of the transmitted signal. [4 marks]

(b) What is the dimensionality D of the signal space? [1 mark]

(c) Plot the constellation diagram. [2 marks]

Answer

(a) $A = 3mV \text{ or } -3mV$

$T_{cs} = 1ms$

$$\text{single pulse} = a.\Lambda\left\{\frac{t}{T_{cs}}\right\} \text{ with } a = \text{random} = \begin{cases} +A \text{ with prob } 0.5 \\ -A \text{ with prob } 0.5 \end{cases}$$

$$\begin{aligned} \text{PSD}(f) &= \frac{1}{T_{cs}}\mathcal{E}\left\{|\text{FT}(\text{single pulse})|^2\right\} = \frac{1}{T_{cs}}\mathcal{E}\left\{\left|\text{FT}\left(a\Lambda\left(\frac{t}{T_{cs}}\right)\right)\right|^2\right\} = \\ &= \frac{1}{T_{cs}}\mathcal{E}\left\{\left|aT_{cs}\text{sinc}^2(fT_{cs})\right|^2\right\} = \frac{1}{T_{cs}}T_{cs}^2\text{sinc}^4(fT_{cs}).\underbrace{\mathcal{E}\{a^2\}}_{(-A)^2\times 0.5 + A^2\times 0.5 = A^2} \\ &= T_{cs}A^2\text{sinc}^2(fT_{cs}^2) \end{aligned}$$

i.e.

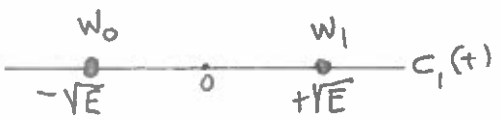
$$\text{PSD}(f) = 9 \times 10^{-9}\text{sinc}^4(f10^{-3}) \text{ W/Hz}$$

(b) $D = 1$;

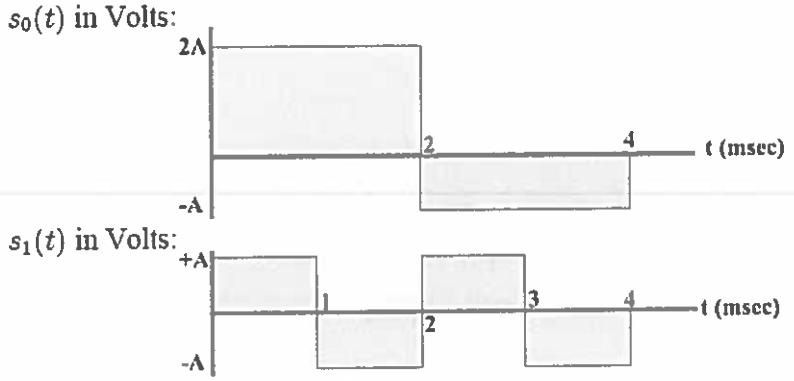
(c) $E = E_0 = E_1 = \frac{1}{3}2T_{cs}A^2 = \frac{1}{3} \times 2 \times 10^{-3} \times 3^2 \times 10^{-6} = 6 \times 10^{-9}$

$w_0 = -\sqrt{E} = -\sqrt{6 \times 10^{-9}}$ and $w_1 = \sqrt{E} = \sqrt{6 \times 10^{-9}}$

$c_1(t) = A\Lambda\left\{\frac{t}{1m}\right\}/\sqrt{E_1} = \frac{3\times 10^{-3}}{\sqrt{6\times 10^{-9}}}\Lambda\left\{\frac{t}{1m}\right\} = 38.730\Lambda\left\{\frac{t}{1m}\right\}$



13. Given a binary digital communication system which employs the following two equally-probable signals and an optimum digital demodulator (matched filter)



The received signals are corrupted by additive white Gaussian channel noise having a double-sided power spectral density of $\frac{7}{8}10^{-9}$ W/Hz.

If the amplitude A is equal to 1mV, find the bit-error rate. [4 marks]

Answer

$p_e = \mathrm{T}\{\sqrt{(1-\rho)\mathrm{EUE}}\}$

$E_b = \frac{1}{2}\int_0^{T_{c*}} \left(s_0(t)^2 + s_1(t)^2\right) dt = \frac{1}{2}\int_0^{2m} \left((1m)^2 + (2m)^2\right) dt + \frac{1}{2}\int_{2m}^{4m} \left((1m)^2 + (1m)^2\right) dt$

$\Rightarrow E_b = \frac{1}{2}(5 \times 2\mu + 2 \times 2\mu) = 7\mu$

$\rho = \frac{1}{E_b}\int_0^{T_c} s_0(t)s_1(t)dt = \ldots\ldots = 0$

$\Rightarrow \rho = 0$

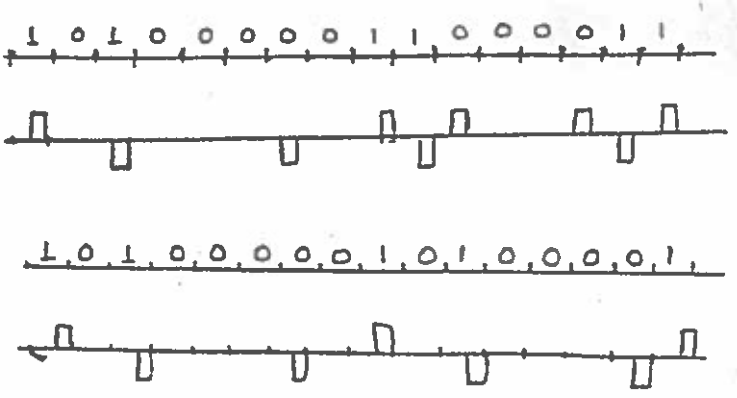
Therefore $\mathrm{EUE} = \frac{7\times10^{-9}}{2\times\frac{7}{8}\times10^{-9}} = 4 \Rightarrow p_e = \mathrm{T}\{\sqrt{4}\} = \mathrm{T}\{2\} = 2.2 \times 10^{-2}$

14. Encode, using HDB3 code, the following two binary sequences [4 marks]

1 0 1 0 0 0 0 0 1 1 0 0 0 0 1 1

1 0 1 0 0 0 0 0 1 0 1 0 0 0 0 1

Answer



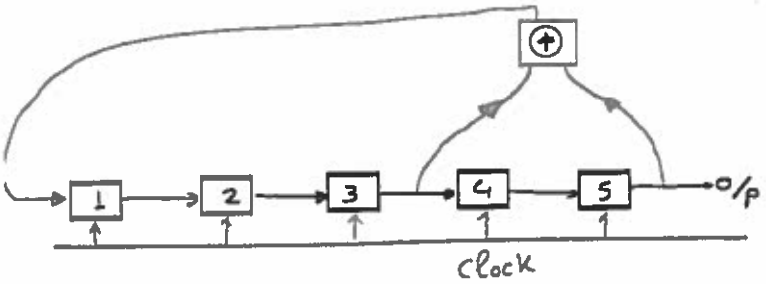
15. Consider a feedback shift register whose feedback connections are represented by the primitive polynomial $D^5 + D^2 + 1$ and operates with a clock rate 2.7Mb/sec.

(a) Sketch this feedback shift register [2 marks]

(b) Find the period of the output sequence in minutes. [1 mark]

(c) Give the first 7 bits of one period of its output sequence - starting with all 1's (initial condition). [2 marks]

Answer



$N = 2^m - 1 \Rightarrow N = 2^5 - 1 = 31$

$m = 5$

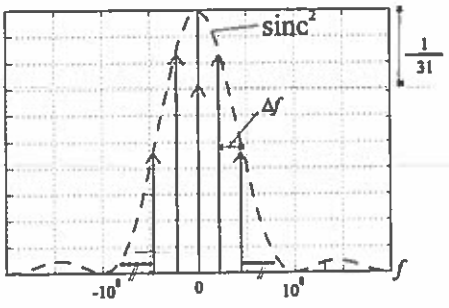
$T_c = \frac{1}{2.7 \times 10^6} = 0.3706\,\mu \frac{\text{sec}}{\text{chip}}$

$NT_c = 31 \times 0.3706\,\mu = 11.4815\,\mu\text{sec} = 0.1914 \times 10^{-6}$ min

	1st	2nd	3rd	4th	o/p 5th
initial condition	1	1	1	1	1
clock pulse No.1	0	1	1	1	1
clock pulse No.2	0	0	1	1	1
clock pulse No.3	0	0	0	1	1
clock pulse No.4	1	0	0	0	1
clock pulse No.5	1	1	0	0	0
clock pulse No.6	0	1	1	0	0
clock pulse No.7	1	0	1	1	0

16. A pseudo random (PN) signal $b(t)$ is generated by using a maximal length shift register of m-stages and has the following double-sided Power Spectral Density.

$\text{PSD}_b(f)$:



(a) Find the number m of shift register stages. [2 marks]

(b) Find Δf. [2 marks]

Answer

(a)

$$\left.\begin{array}{l} \frac{1}{T_c} = 10M \Rightarrow T_c = 10^{-7}\,\text{sec} \\ \frac{1}{N_c T_c} = 39.2k \Rightarrow N_c T_c = 25.5\mu\,\text{sec} \end{array}\right\} \Rightarrow \quad N_c = 255$$

$$N_c = 2^m - 1 \Rightarrow \qquad m = \log_2(N_c + 1) = 8$$

(b) $\Delta f = \frac{1}{N_c T_c} = 39.2k$

17. Consider a PCM system where its sampler operates at twice the Nyquist rate and its quantizer consists of a μ-law compander (with $\mu = 100$) followed by a Q-level uniform quantizer.
Note that μ-law compression is defined as follows:

$$\text{output} = \frac{\ln(1+\mu.\,|\,x\,|)}{\ln(1+\mu)}\text{sign}(x) \qquad \text{where } x = \frac{\text{input value in } Volts}{\text{maximum input value in } Volts}$$

The maximum and minimum values of the input signal are +5Volts and −5Volts respectively and the signal occupies a bandwidth from 100Hz to 15kHz. It is required that the Signal-to-Quantization-Noise Ratio (SNRq) should be better than 8.5 dB

(a) Find the minimum number of quantizing levels Q required, as well as the transmission bit rate. [2 marks]

(b) If the signal at the output of the sampler at time kT_s is equal to −2.4 Volts, estimate the instantanous quantization noise $n_q(kT_s)$ where T_s is the sampling period and k is an integer. [4 marks]

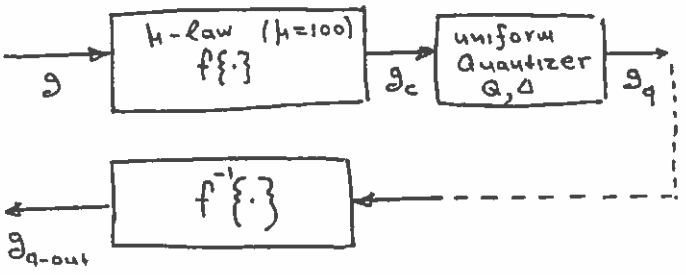
Answer

(1) $\text{SNR}_q \geqslant 8.5\text{dB}$, $\mu = 100$

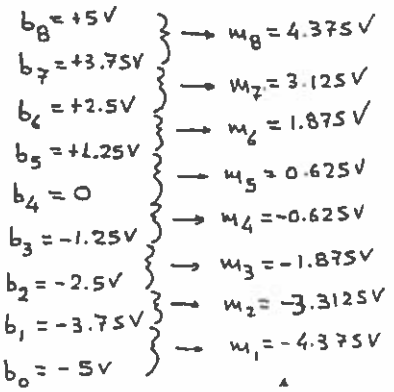
$$4.77+6\gamma - \underbrace{20\log_{10}(\ln(1+\mu))}_{13.2837} \geqslant 8.5$$

$\gamma \geq 2.8356 \Rightarrow \gamma = 3 \Rightarrow Q = 2^\gamma = 8$ levels

$r_b = \gamma \times 2 \times 2 \times 15k = 180$ kbits/sec



(2)



$$\left|\frac{g}{g_{max}}\right| = \left|\frac{-2.4}{5}\right| = 0.48$$

$$|g_c| = \frac{\ln(1+100\times 0.48)}{\ln(1+100)} \times \underbrace{g_{\max}}_{5} = 4.2164 \text{ V}$$

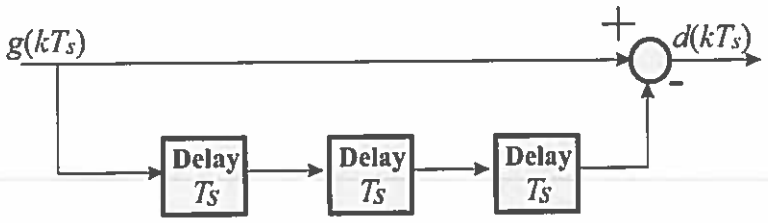
$\Rightarrow b_0 < g_c < b_1 \Rightarrow g_q = -4.375V = m_1$

$|g_{q,out}| = \frac{1}{\mu}\left(\exp\left(\frac{m_1}{g_{max}} \times \ln(1+\mu)\right) - 1\right) \times 5 = \frac{1}{\mu}(\exp(4.0382) - 1) \times 5 = 2.7863$

$g_{q,out} = -2.7863V$

$n_q = |-2.4 + 2.7863| = 0.3863$

18. Consider a non-white zero-mean Gaussian random source $g(t)$ having an autocorrelation function $R_{gg}(\tau) = \exp(-8000 \mid \tau \mid)$. The signal is sampled at a rate of 24 ksamples/sec and then is applied at the input the following circuit



For the signal $d(kT_s)$, where T_s is the sample time, k is an integer, find:

(a) the *DC* level, [1 mark]

(b) the average power. [3 marks]

Answer

- $d(kT_s)$ is a Gaussian signal i.e. $N(\mu_d, \sigma_d)$
  mean: $\mu_d = E\{d(kT_s)\} = E\{g(kT_s) - g((k-3)T_s)\} = 0 - 0 = 0$
  std: $\sigma_d = \sqrt{P_d}$
  where $P_d = E\{d^2(kT_s)\} = E\left\{(g(kT_s) - g((k-3)T_s))^2\right\} =$
  $= E\{g^2(kT_s)\} + E\{g^2((k-3)T_s)\} - 2E\{g(kT_s)g((k-3)T_s)\}$
  $= R_{gg}(0) + R_{gg}(0) - 2R_{gg}(3T_s) = 2R_{gg}(0) - 2R_{gg}(3T_s)$ where $T_s = \frac{1}{24k}$
  $= 2 \times \exp(-8000 \mid 0 \mid) - 2 \times \exp(-8000 \mid \frac{3}{24k} \mid)$
  $= 2 - 2 \times \exp(-1) = 1.264$
  i.e. pdf=$N(0, \sqrt{1.264})$

19. A high quality music signal with a Crest Factor of 4.4668, having a maximum frequency 18 kHz, is applied to a uniform PCM system (i.e. PCM with a uniform quantizer). If it is specified that the Signal-to-Quantisation-Noise ratio (SNR_q) should be better than 50dB, find the minimum data bit rate required [6 marks]

Answer

$\text{SNR}_q \geqslant 50\ dB; CR = \frac{\hat{V}}{\sigma} = 4.4668$

$$\begin{aligned} \text{SNR}_q &= 4.77 + 6\gamma - \underbrace{20\log_{10}\frac{\hat{V}}{\sigma}}_{=13dB}\ \text{dB} \\ \text{SNR}_q &= 4.77 + 6\gamma - 13 \end{aligned}$$

i.e.

$$\begin{aligned} &\Rightarrow\ 4.77 + 6\gamma - 13 \geqslant 50 \\ &\Rightarrow\ \gamma \geqslant \frac{50 + 13 - 4.77}{6} \\ &\Rightarrow\ \gamma \geqslant 10 \Rightarrow \log_2 Q \geqslant 10 \Rightarrow Q \geqslant 2^{10} = 1024 \end{aligned}$$

Therefore,

$$r_b = \gamma F_s = \gamma 2 F_g = 10 \times 2 \times 18k = 360 kbits/\sec$$

20. In a binary PCM communication system prove that bandwidth expansion factor β is equal to the average number of bits γ per quantization level, i.e. [4 marks]

$$\beta = \gamma$$

Answer

$B = \frac{\text{channel symbol rate}}{2} = \frac{\text{bit rate}}{2} = \frac{\gamma F_s}{2} = \gamma F_g \Rightarrow B = \gamma F_g \Rightarrow \frac{B}{F_g} = \gamma \Rightarrow \beta = \gamma$

21. The two signals $s_0(t)$ and $s_1(t)$ of a binary communication system each have energy equal to 93.3 and cross correlation coefficient 0.866.

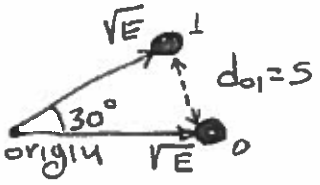
(a) Draw the constellation diagram of the system properly labeled. [1 mark]

(b) What is the distance of these two signals? [3 marks]

Answer

(a) $\rho_{01} = \cos\phi \Rightarrow 0.866 = \cos\phi \Rightarrow \phi = 30°$

(b) $d_{01} = 2E - 2 \times 0.866 \times \sqrt{E \times E} \Rightarrow d_{01}^2 = 25 \Rightarrow d_{01} = 5$



22. Define the terms in a CDMA wireless channel:

- delay spread, coherence bandwidth and flat fading . [3 marks]
- channel selectivity and channel coherence [2 marks]

Answer

delay spread : is the time it takes for light to travel a distance equal to the longest path minus the shortest path

i.e. $\frac{\max_{\forall j}(d_j) - \min_{\forall j}(d_j)}{c}$

coherent bandwidth $\triangleq \frac{1}{\text{delay spread}}$

flat fading : the CDMA receiver is unable to separate the multipath signals if the delay between them is less that the period of a chip T_c

- Channel *Selectivity*: A channel has selectivity if it varies as a function of either time, frequency, or space
  Channel *Coherence*: (opposite of Channel Selectivity): A channel has coherence if it does not vary as a function of either time, frequency, or space over a specified 'window' of interest.
  This is the **most important** concept in describing wireless channels
- coherence: { *temporal* coherence -coherence time T_{coh} ; *frequency* coherence -coherence bandwidth B_{coh} ; *spatial* coherence -coherence distance D_{coh} }

23. Describe the log-distance path loss model of a wireless channel. [3 marks]

Answer

$$\begin{aligned}\text{Path-Loss(PL)} &= \underbrace{10\log_{10}\frac{\overset{Watt}{\downarrow}P_{Tx}}{1mW}}_{Tx \text{ power in dBm}} - \underbrace{10\log_{10}\frac{\overset{Watt}{\downarrow}P_{Rx}}{1mW}}_{Rx \text{ power in dBm}} \text{ (dB)} \\ &= \text{PL}_0 + 10\log_{10}\left(\frac{d}{d_0}\right)^a + \text{PL}_{Gaussian} \text{ (dB)}\end{aligned}$$

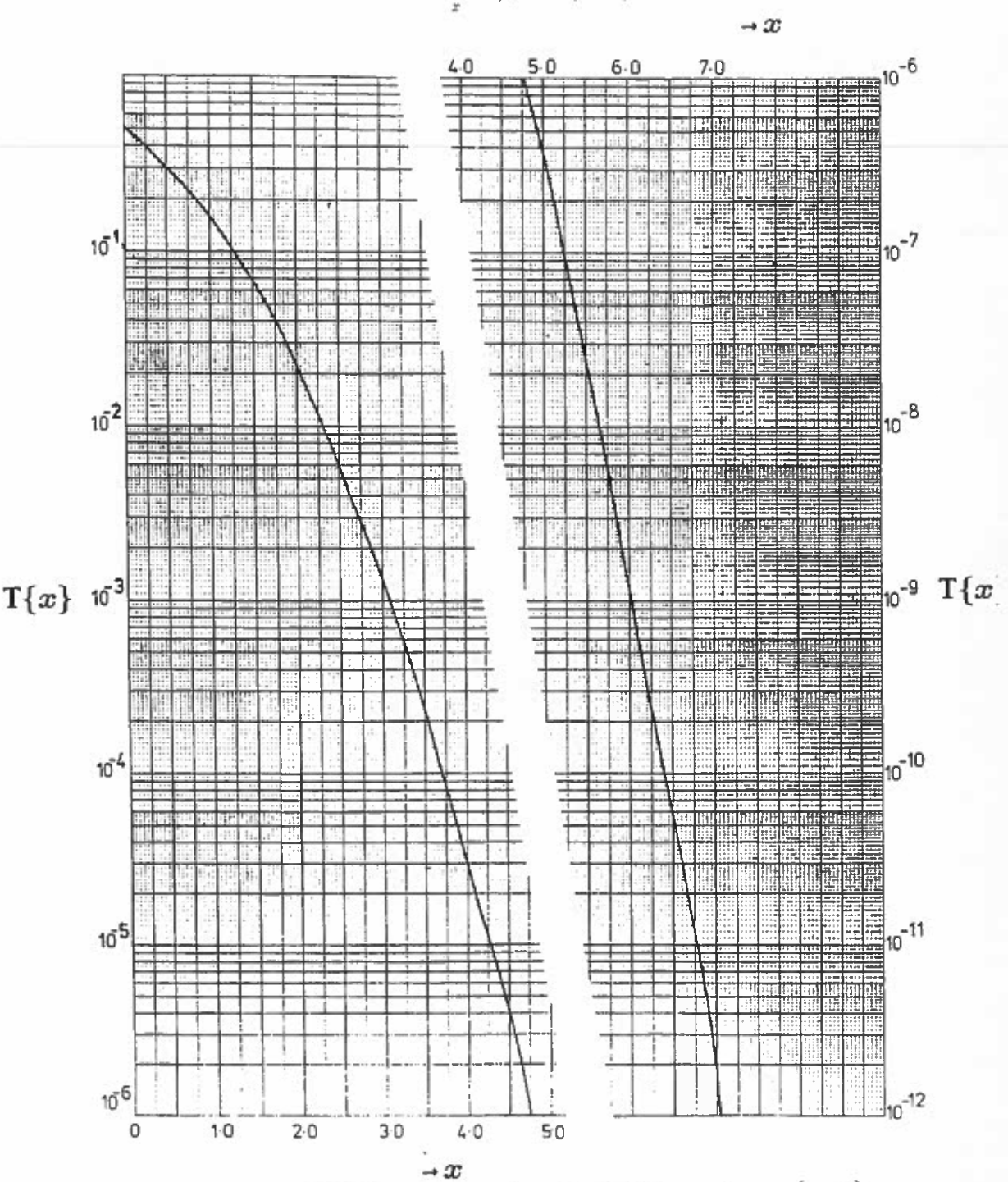
where

PL_0 = the path loss at the reference distance $d_0 = 1km/1mile$
d = path length
a = path loss exponent
$\text{PL}_{Gaussian}$ = $N(0, \sigma^2)$.

Fourier Transform Tables			
	Description	Function	Transformation
1	Definition	$g(t)$	$G(f) = \int_{-\infty}^{\infty} g(t).e^{-j2\pi ft}dt$
2	Scaling	$g\left(\frac{t}{T}\right)$	$\|T\|.G(fT)$
3	Time shift	$g(t-T)$	$G(f).e^{-j2\pi fT}$
4	Frequency shift	$g(t).e^{j2\pi Ft}$	$G(f-F)$
5	Complex conjugate	$g^*(t)$	$G^*(-f)$
6	Temporal derivative	$\frac{d^n}{dt^n}g(t)$	$(j2\pi f)^n.G(f)$
7	Spectral derivative	$(-j2\pi t)^n.g(t)$	$\frac{d^n}{df^n}G(f)$
8	Reciprocity	$G(t)$	$g(-f)$
9	Linearity	$A.g(t) + B.h(t)$	$A.G(f) + B.H(f)$
10	Multiplication	$g(t).h(t)$	$G(f) * H(f)$
11	Convolution	$g(t) * h(t)$	$G(f).H(f)$
12	Delta function	$\delta(t)$	1
13	Constant	1	$\delta(f)$
14	Rectangular function	$\mathbf{rect}\{t\} \triangleq \begin{cases} 1 & \text{if } \|t\| < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$	$\mathbf{sinc}\{f\} \triangleq \frac{\sin(\pi f)}{\pi f}$
15	Sinc function	$\mathbf{sinc}(t)$	$\mathbf{rect}\{f\}$
16	Unit step function	$u(t) \triangleq \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$	$\frac{1}{2}\delta(f) - \frac{j}{2\pi f}$
17	Signum function	$\mathrm{sgn}(t) \triangleq \begin{cases} 1 & t > 0 \\ -1 & t < 0 \end{cases}$	$-\frac{j}{\pi f}$
18	decaying exp (two-sided)	$e^{-\|t\|}$	$\frac{2}{1+(2\pi f)^2}$
19	decaying exp (one-sided)	$e^{-\|t\|}.u(t)$	$\frac{1-j2\pi f}{1+(2\pi f)^2}$
20	Gaussian function	$e^{-\pi t^2}$	$e^{-\pi f^2}$
21	Lambda function	$\Lambda\{t\} \triangleq \begin{cases} 1-t & \text{if } 0 \le t \le 1 \\ 1+t & \text{if } -1 \le t \le 0 \end{cases}$	$\mathbf{sinc}^2\{f\}$
22	Repeated function	$\mathbf{rep}_T\{g(t)\} = g(t) * \mathbf{rep}_T\{\delta(t)\}$	$\left\|\frac{1}{T}\right\|\mathbf{comb}_{\frac{1}{T}}\{G(f)\}$
23	Sampled function	$\mathbf{comb}_T\{g(t)\} = g(t).\mathbf{rep}_T\{\delta(t)\}$	$\left\|\frac{1}{T}\right\|\mathbf{rep}_{\frac{1}{T}}\{G(f)\}$

The graph below shows the Tail function $\mathbf{T}\{x\}$ which represents the area from x to ∞ of the Gaussian probability density function N(0,1), i.e.

$$\mathbf{T}\{x\} = \int_x^{\infty} \frac{1}{\sqrt{2\pi}} . \exp\left(-\frac{y^2}{2}\right) dy$$



Note that if $x > 6.5$ then $\mathbf{T}\{x\}$ may be approximated by $\mathbf{T}\{x\} \approx \frac{1}{\sqrt{2\pi}.x} . \exp\left\{-\frac{x^2}{2}\right\}$